VERTICAL TOWER CULTIVATION SYSTEM

BSIT-S-2A

Group 6 - Computer Organization

MEMBERS:

TOMON, JUSTINE O.
PEQUE, ERICA M.
GARCIA, AIA A.

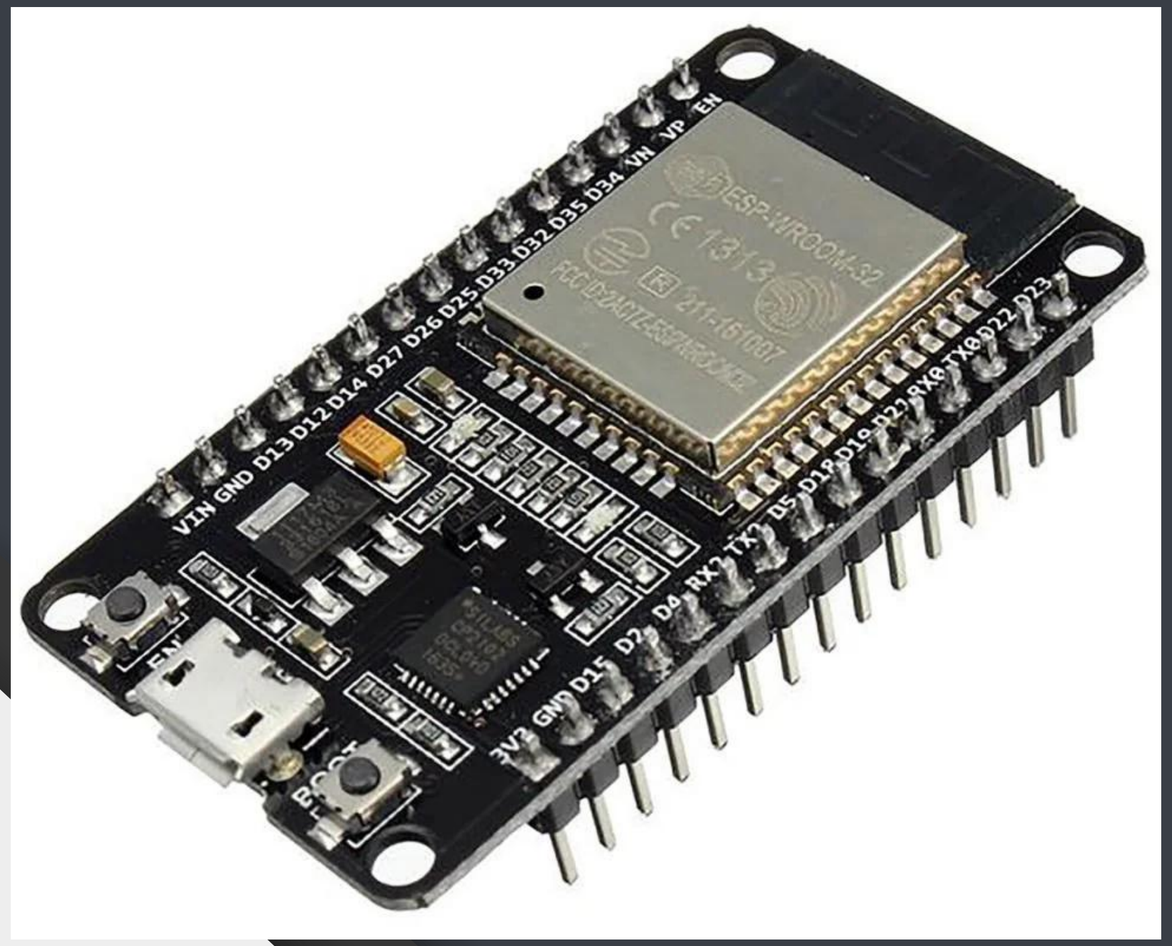
INTRODUCTION

The Vertical Tower Cultivation System aims to improve basil cultivation by using automatic control systems which manage watering as well as optimize light exposure and track plant growth. The system employs advanced sensors together with responsive actuators to preserve ideal basil cultivation environments within its compact vertical structure. The automated system both optimizes plant cultivation operations and establishes perfect growing conditions for basil cultivation indoors.

OBJECTIVES

1. The automated watering system will ensure moisture consistency through water pump control based on soil condition monitoring.
2. A lighting system should activate LED grow lights based on surrounding light conditions to help basil grow successfully and reduce power consumption.
3. An automated humidifier will guarantee ideal humidity conditions for basil growth.
4. The system uses sensors to track environmental factors as they enable scientific plant care by collecting data.
5. The system needs a web-based platform for system monitoring functions.

MATERIALS



ESP32

A Wi-Fi and Bluetooth-enabled microcontroller for controlling sensors, relays, and automation.

MATERIALS



DC DC Converter

Converts voltage (e.g., 12V/24V to 5V) to power the ESP32 and sensors.

MATERIALS



4 Channel Relay

Controls up to 4 devices independently.

MATERIALS - SENSORS



CAPACITIVE SOIL MOISTURE

Measures soil moisture without direct contact, providing an analog signal.

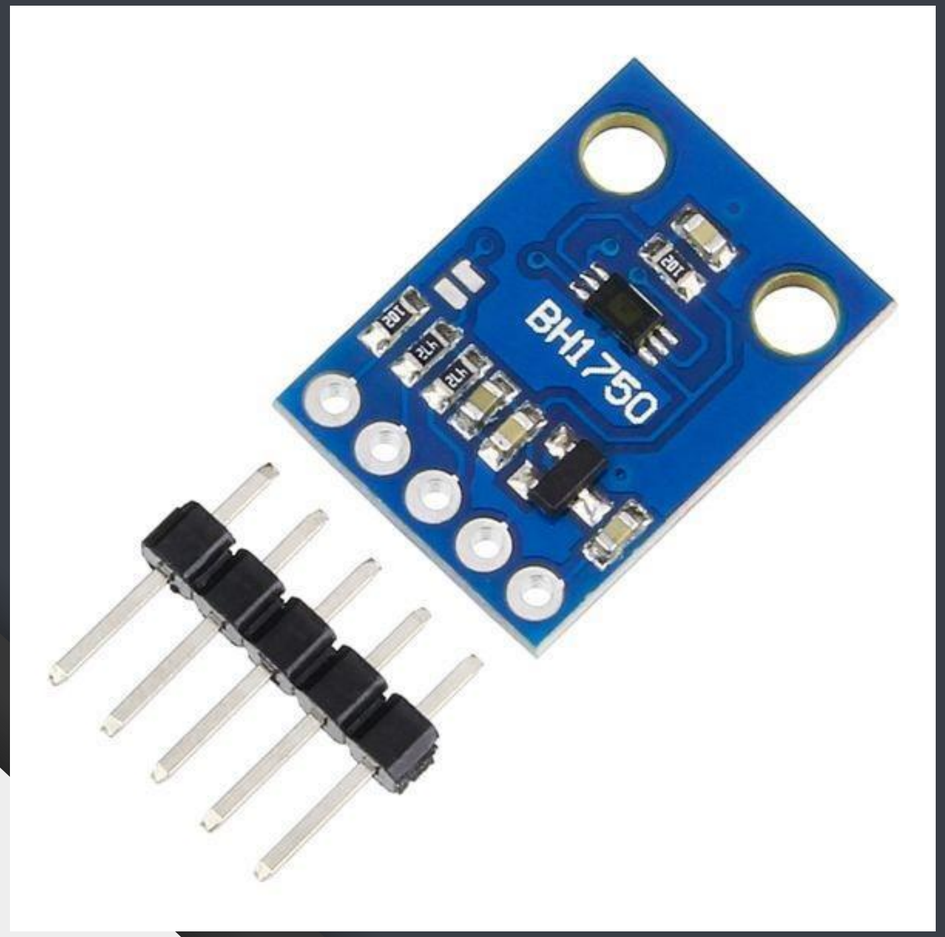
MATERIALS - SENSORS



DHT22

A digital temperature and humidity sensor with high accuracy.

MATERIALS - SENSORS



BH1750

An ambient light sensor that measures light intensity in lux via I2C.

MATERIALS - OUTPUTS



WATER PUMP

A 5v pump that moves water to plants, controlled by a relay.

MATERIALS - OUTPUTS



LED Grow Light

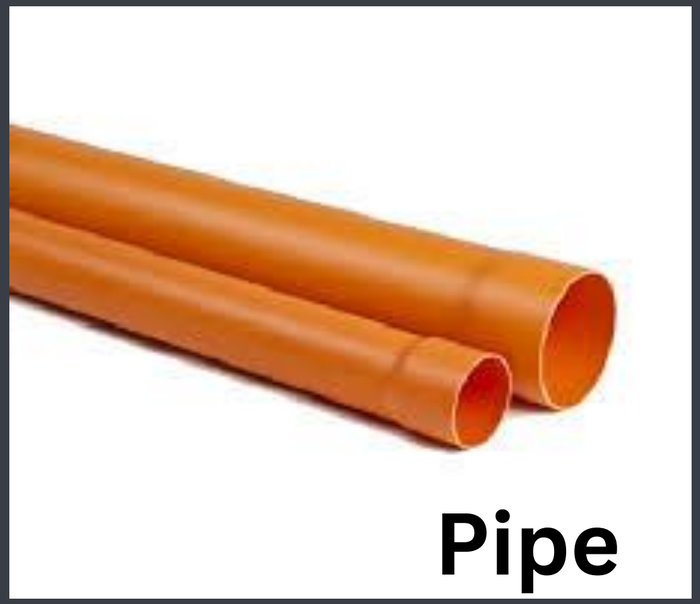
A specialized LED light that emits the optimal spectrum for plant growth, controlled by a relay.

MATERIALS - OUTPUTS



Humidifier

A device that increases air humidity by releasing mist, controlled based on humidity levels.

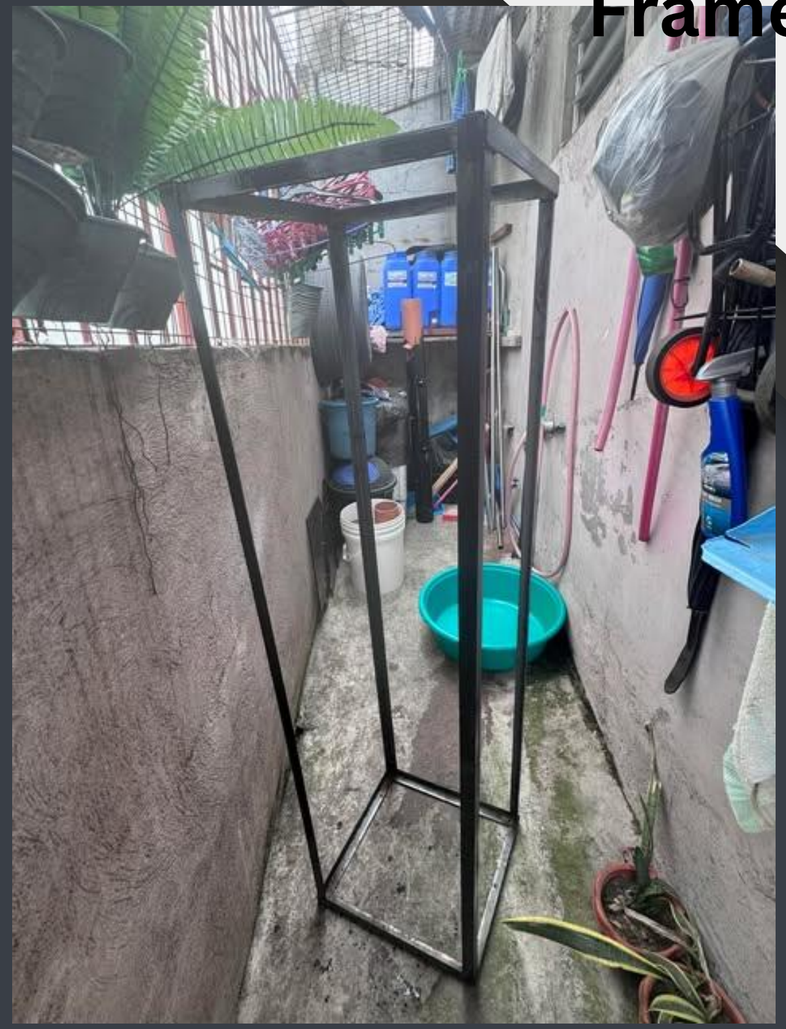


Pipe



Pai
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Metal
Frame

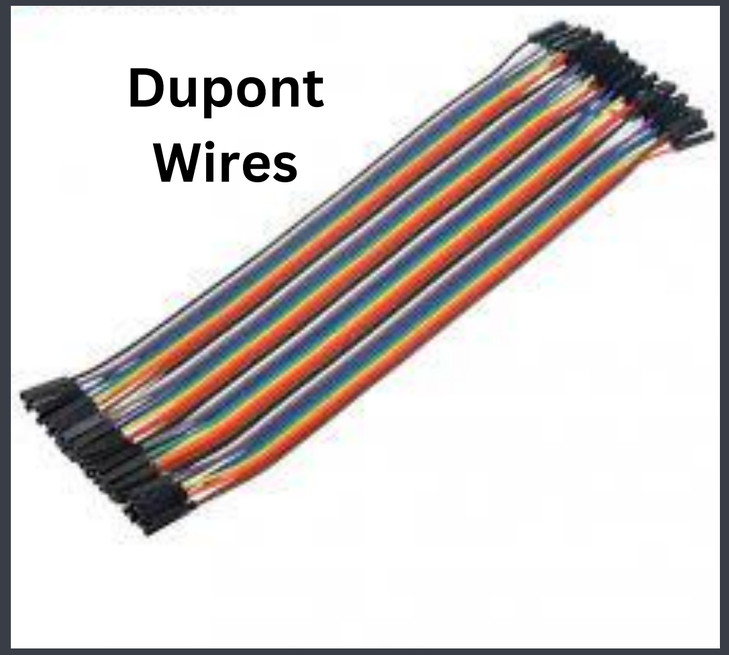


STRUCTURAL MATERIALS

ADDITIONAL MATERIALS



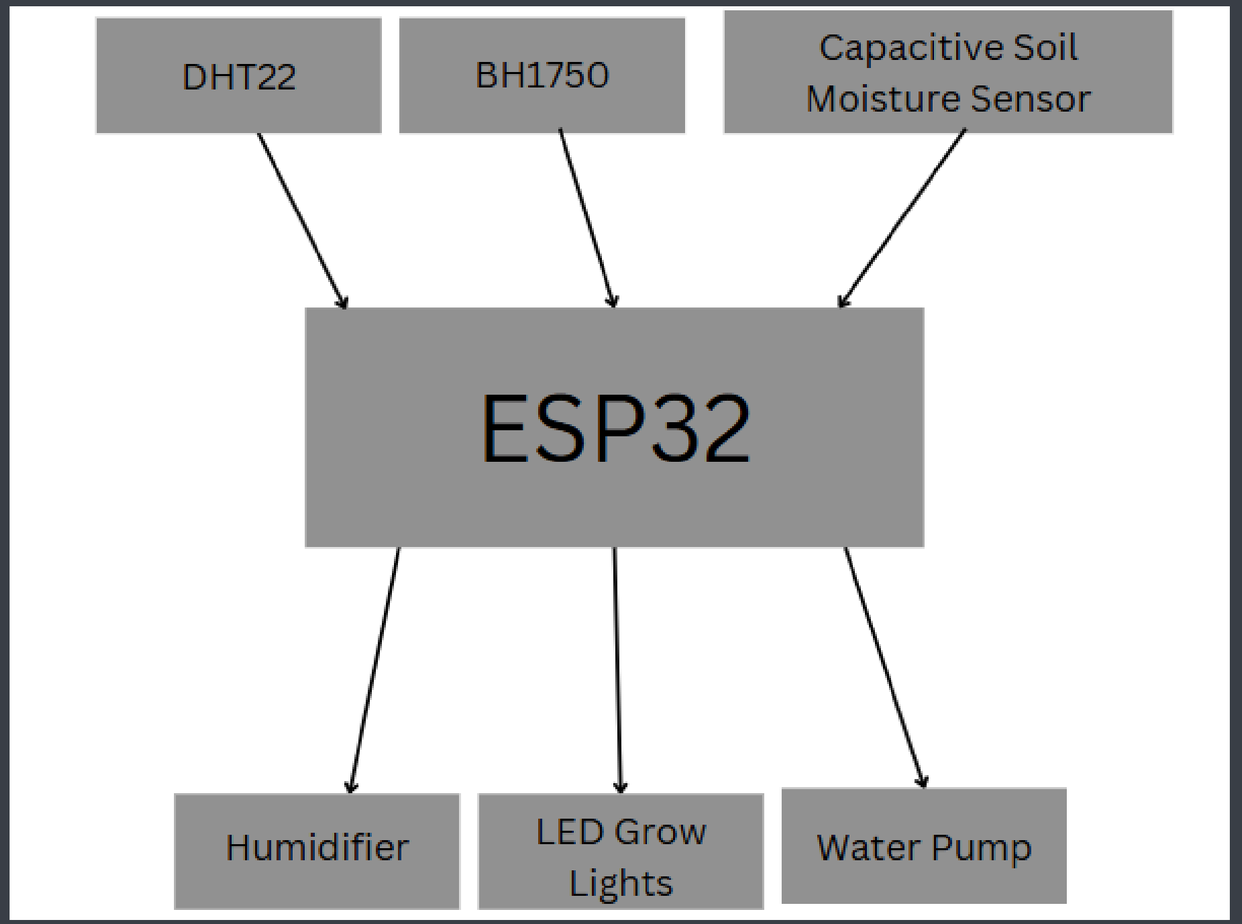




INPUT SENSORS

- Soil Moisture Sensor (Capacitive - v1.2): Measures soil moisture based on capacitance. Outputs an analog signal where higher values indicate dry soil and lower values indicate wet soil.
- DHT22 (Humidity and Temperature Sensor): Measures humidity (%) and temperature (°C) using a digital interface.
- BH1750 (Light Sensor): Measures ambient light intensity in lux using an I2C interface.

BLOCK DIAGRAM

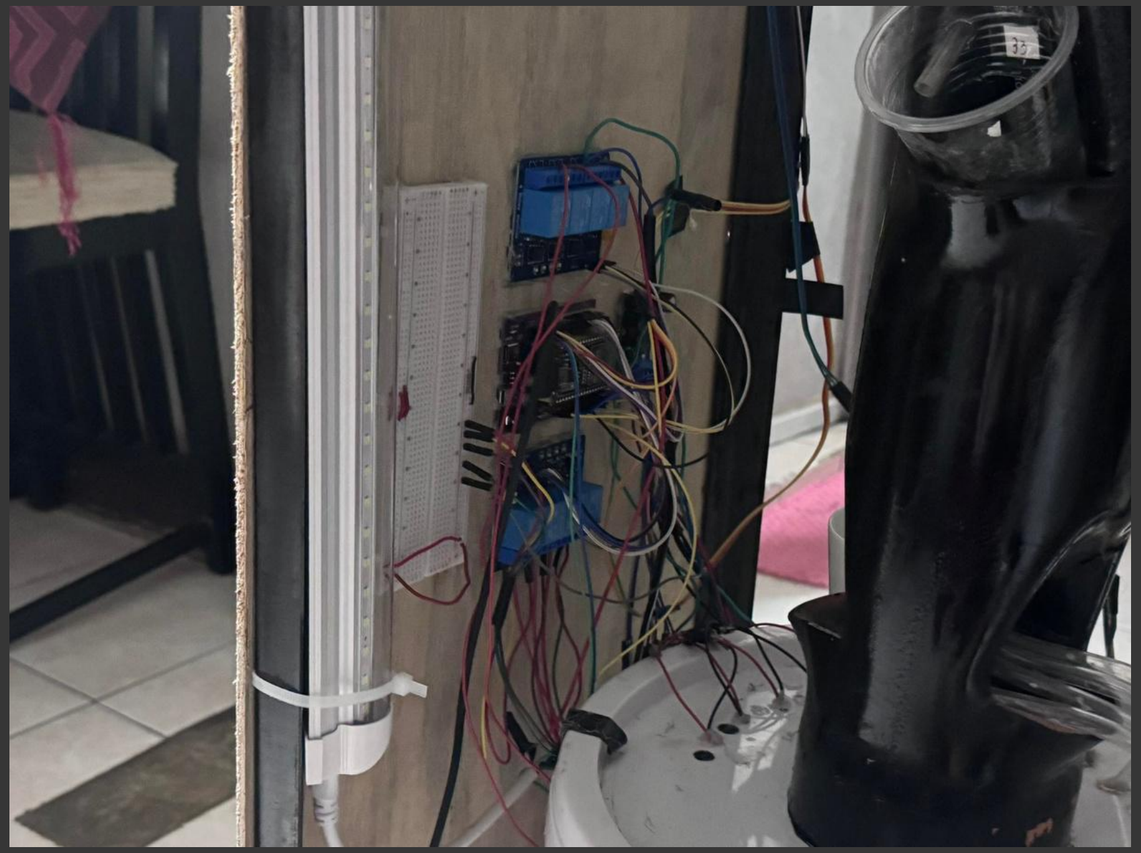


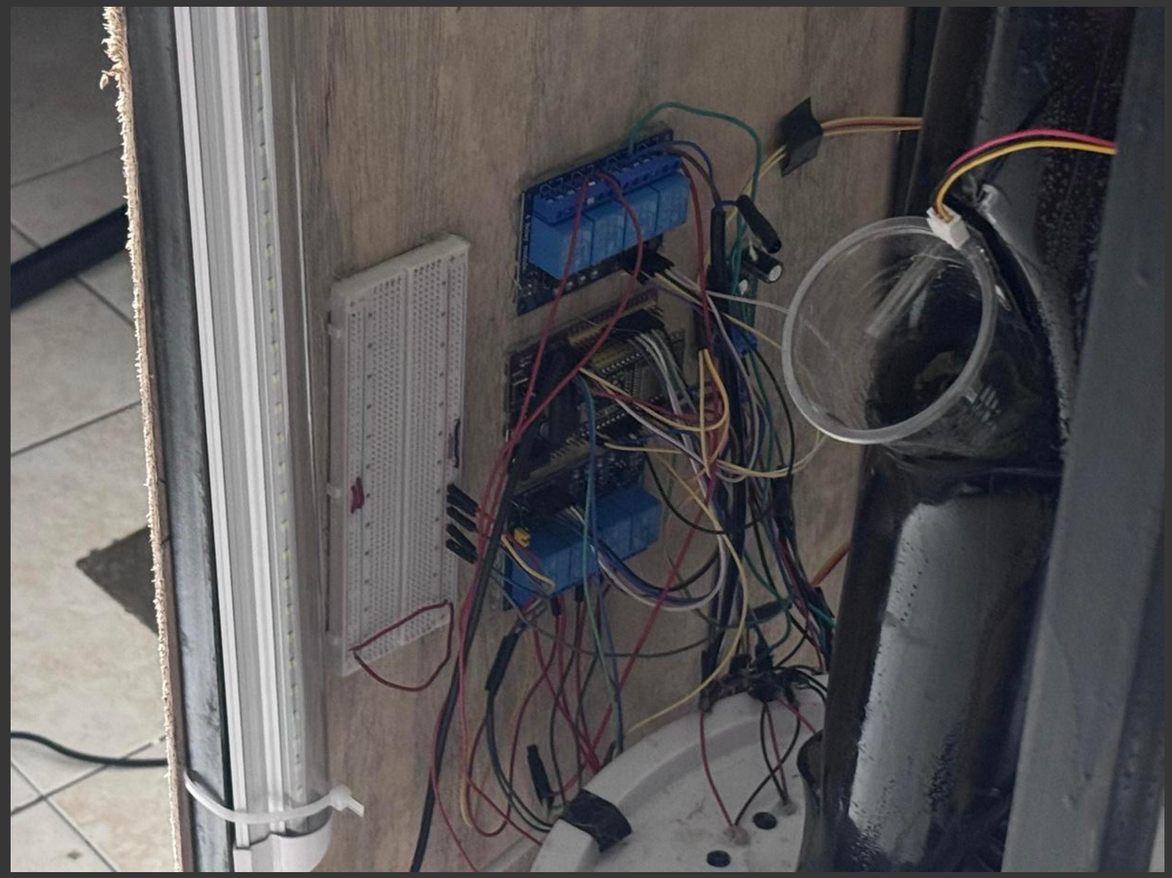
PIN CONFIGURATIONS

Component	ESP32 Pin
DHT22 (Temperature & Humidity)	GPIO 17
BH1750 (Light Sensor)	GPIO 21 (SDA), GPIO 22 (SCL)
Humidifier Relay	GPIO 15
Light Relay 1	GPIO 19
Light Relay 2	GPIO 5
Water Pump Relay 1	GPIO 25
Water Pump Relay 2	GPIO 26
Water Pump Relay 3	GPIO 4
Water Pump Relay 4	GPIO 14

Soil Moisture Sensor 1	GPIO 34 (Analog)
Soil Moisture Sensor 2	GPIO 35 (Analog)
Soil Moisture Sensor 3	GPIO 32 (Analog)
Soil Moisture Sensor 4	GPIO 33 (Analog)

WIRING







FINAL OUTPUT

DOCUMENTATION







DOCUMENTATION





DOCUMENTATION





DOCUMENTATION





DOCUMENTATION







REFERENCES

Nor, M. S. a. M., Sani, M. a. A., Yusof, M. I., Asari, A. R., Hassan, S., & Sani, N. S. (2023). Development of vertical indoor farming using an arduino and IoT. In SpringerBriefs in applied
sciences and technology (pp. 79-85). https://doi.org/10.1007/978-3-031-47727-0_10

Hackster.io. (2023, August 20). Automated vertical farming.
https://www.hackster.io/go-b/automated-vertical-farming-72737c

Iseman, L. (2022, September 30). Garduino: Geek Gardening with Arduino. Make: DIY Projects
and Ideas for Makers. https://makezine.com/projects/garduino-geek-gardening/

Journal, I. (2023, January 1). Smart Cultivation: an arduino-based IoT aeroponics system for indoor farming.
https://www.academia.edu/107406764/Smart_Cultivation_An_Arduino_based_IoT_Aeroponics_
System_for_Indoor_Farming

THANK YOU

FOR LISTENING